

The Pythagorean Theorem: A Comprehensive Guide

The Pythagorean theorem is a fundamental principle in geometry that describes the relationship between the lengths of the sides in a right-angled triangle. It is one of the most famous and widely used mathematical formulas in the world.

Definition

The **Pythagorean theorem** states that in any right-angled triangle, the area of the square whose side is the hypotenuse (the side opposite the right angle) is equal to the sum of the areas of the squares whose sides are the two legs.

Understanding the Components

To apply the theorem, we must first identify the parts of a right triangle. A right triangle is defined by having one angle that is exactly 90 degrees.

- **Legs (a and b):** These are the two shorter sides that meet to form the right angle.
- **Hypotenuse (c):** This is the longest side of the triangle, located directly across from the right angle.

The Fundamental Formula

The relationship between these sides is expressed by the following equation:

$$a^2 + b^2 = c^2$$

This means that if you know the lengths of any two sides of a right triangle, you can calculate the length of the third side using basic algebra.

Worked Example: Finding the Hypotenuse

Before we conclude, let's see the theorem in action. Imagine you have a right triangle where the two legs measure 3 units and 4 units. We want to find the length of the hypotenuse.

Example

Problem: Given $a = 3$ and $b = 4$, find c .

Step 1: Plug the values into the formula.

$$3^2 + 4^2 = c^2$$

Step 2: Square the numbers.

$$9 + 16 = c^2$$

Step 3: Add the results.

$$25 = c^2$$

Step 4: Take the square root of both sides.

$$\sqrt{25} = c$$

Result: $c = 5$

Important Takeaways

- The theorem **only** applies to right-angled triangles.
- The hypotenuse c is always the longest side.
- A set of three integers that satisfy the theorem (like 3, 4, 5) is called a **Pythagorean Triple**.

I have finished generating your one-page note on the Pythagorean theorem. You can now use this as a quick reference guide for your studies.